

## Lecture: Nios II Software

Lecturer: Dr. Nguyen Anh Vinh

Ref: Truong Quang Vinh, IT-SoC Lab

### ❖ **Class content**

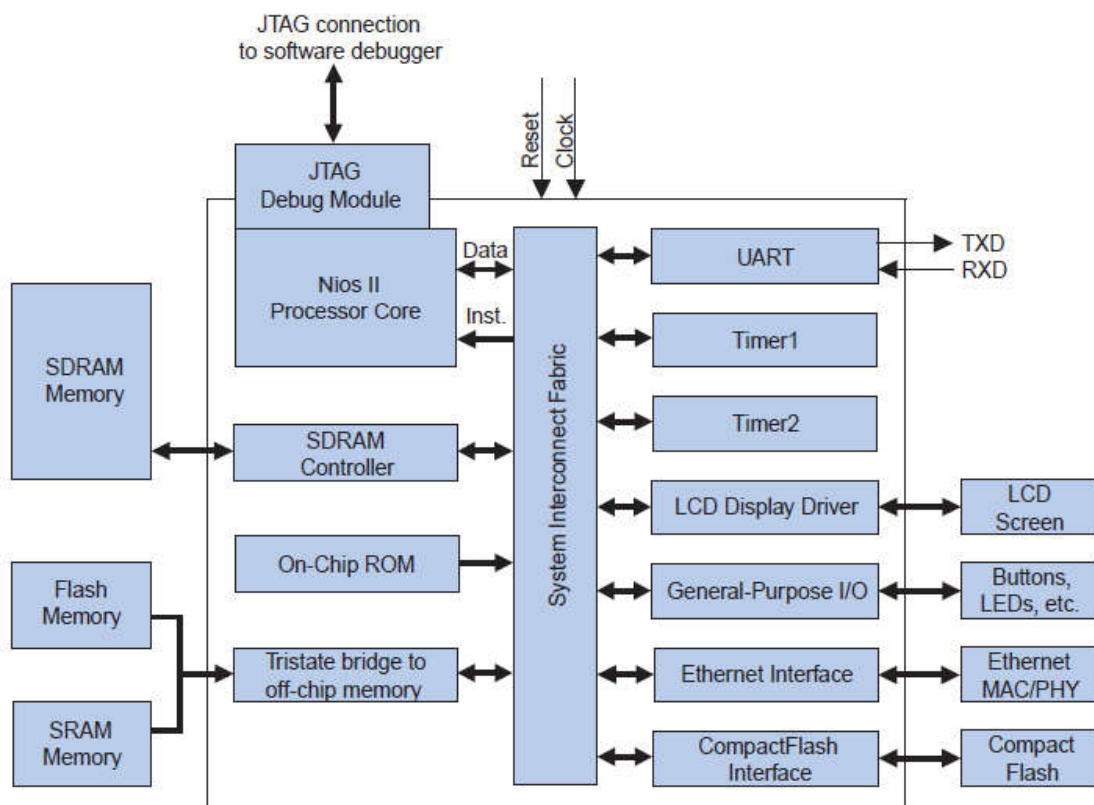
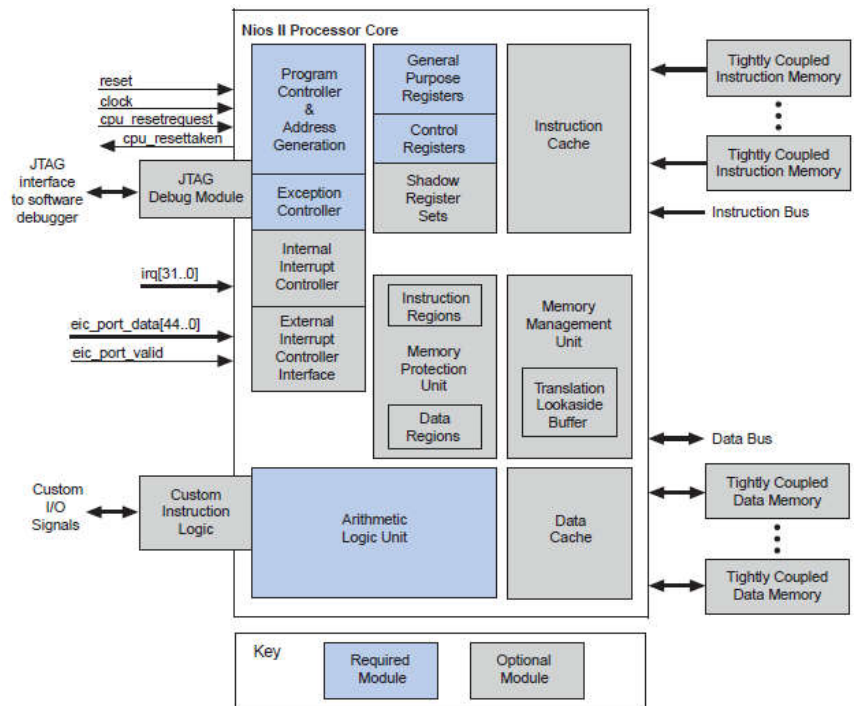
- Nios II System Development Flow
- Avalon-ST interface
- Conduit interface

### ❖ **Required documents**

- Nios II Hardware Development Tutorial
- Nios II Processor Reference Handbook

## ❖ The Nios II processor is a general-purpose RISC processor core

- Full 32-bit instruction set, data path, and address space
- Software development environment based on the GNU C/C++ tool chain
- Performance up to 250 DMIPS (Dhrystone MIPS)

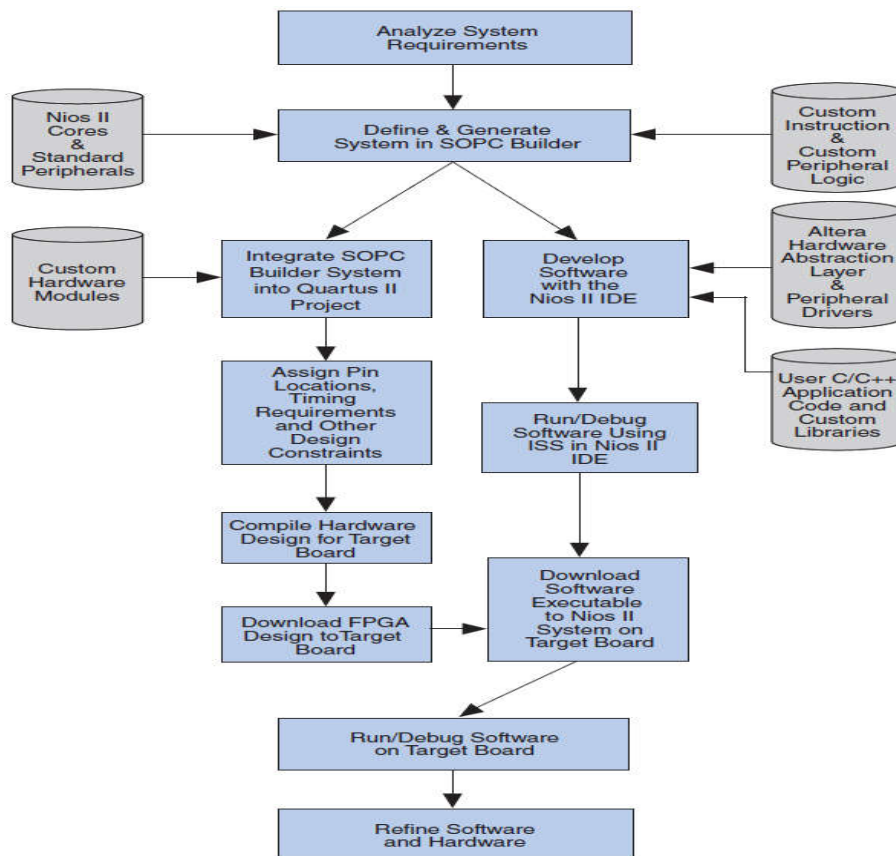


**Table 5–1. Nios II Processor Cores (Part 1 of 3)**

| Feature     |                           | Core                     |                           |                                                                                                                                        |
|-------------|---------------------------|--------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|             |                           | Nios II/e                | Nios II/s                 | Nios II/t                                                                                                                              |
| Objective   |                           | Minimal core size        | Small core size           | Fast execution speed                                                                                                                   |
| Performance | DMIPS/MHz (1)             | 0.15                     | 0.74                      | 1.16                                                                                                                                   |
|             | Max. DMIPS (2)            | 31                       | 127                       | 218                                                                                                                                    |
|             | Max. $f_{\text{MAX}}$ (2) | 200 MHz                  | 165 MHz                   | 185 MHz                                                                                                                                |
| Area        |                           | < 700 LEs;<br>< 350 ALMs | < 1400 LEs;<br>< 700 ALMs | Without MMU or MPU:<br>< 1800 LEs;<br>< 900 ALMs<br>With MMU:<br>< 3000 LEs;<br>< 1500 ALMs<br>With MPU:<br>< 2400 LEs;<br>< 1200 ALMs |
| Pipeline    |                           | 1 stage                  | 5 stages                  | 6 stages                                                                                                                               |

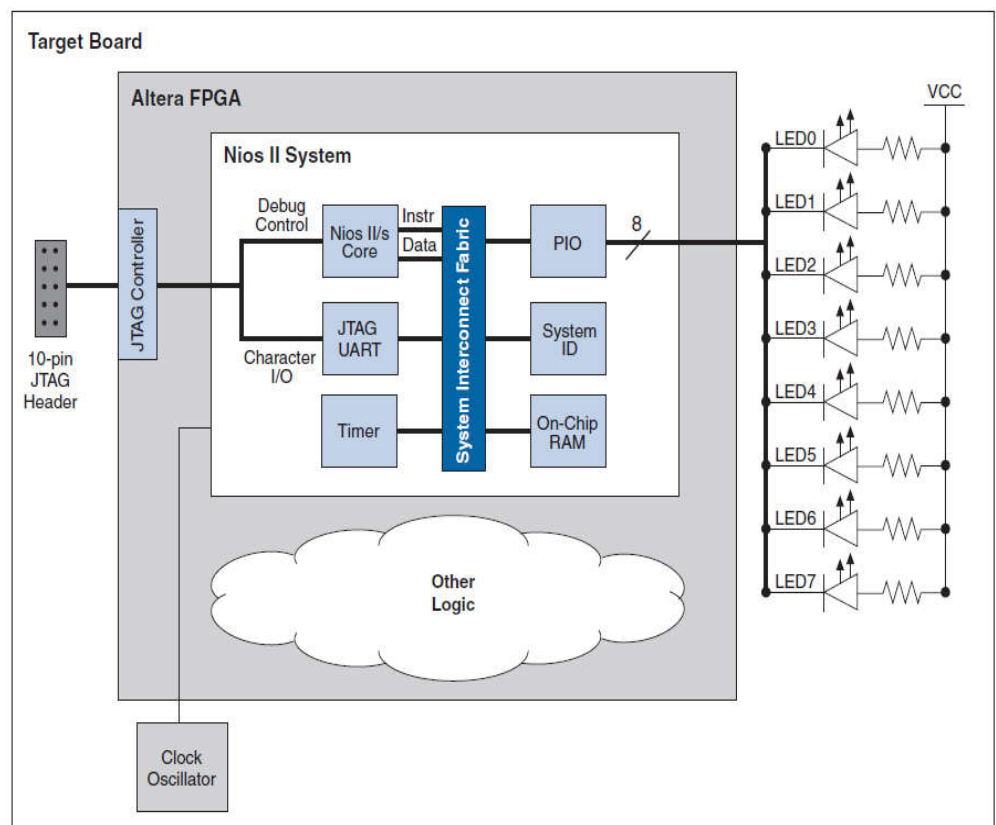
**Table 7–1. Representation of Data Types**

| Type                | Size (Bytes) | Representation           |
|---------------------|--------------|--------------------------|
| char, signed char   | 1            | two's complement (ASCII) |
| unsigned char       | 1            | binary (ASCII)           |
| short, signed short | 2            | two's complement         |
| unsigned short      | 2            | binary                   |
| int, signed int     | 4            | two's complement         |
| unsigned int        | 4            | binary                   |
| long, signed long   | 4            | two's complement         |
| unsigned long       | 4            | binary                   |
| float               | 4            | IEEE                     |
| double              | 8            | IEEE                     |
| pointer             | 4            | binary                   |
| long long           | 8            | two's complement         |
| unsigned long long  | 8            | binary                   |



### ❖ Nios II system:

- Nios II processor
- Onchip memory
- Timer
- JTAG UART
- 8-bit parallel I/O
- System identification component



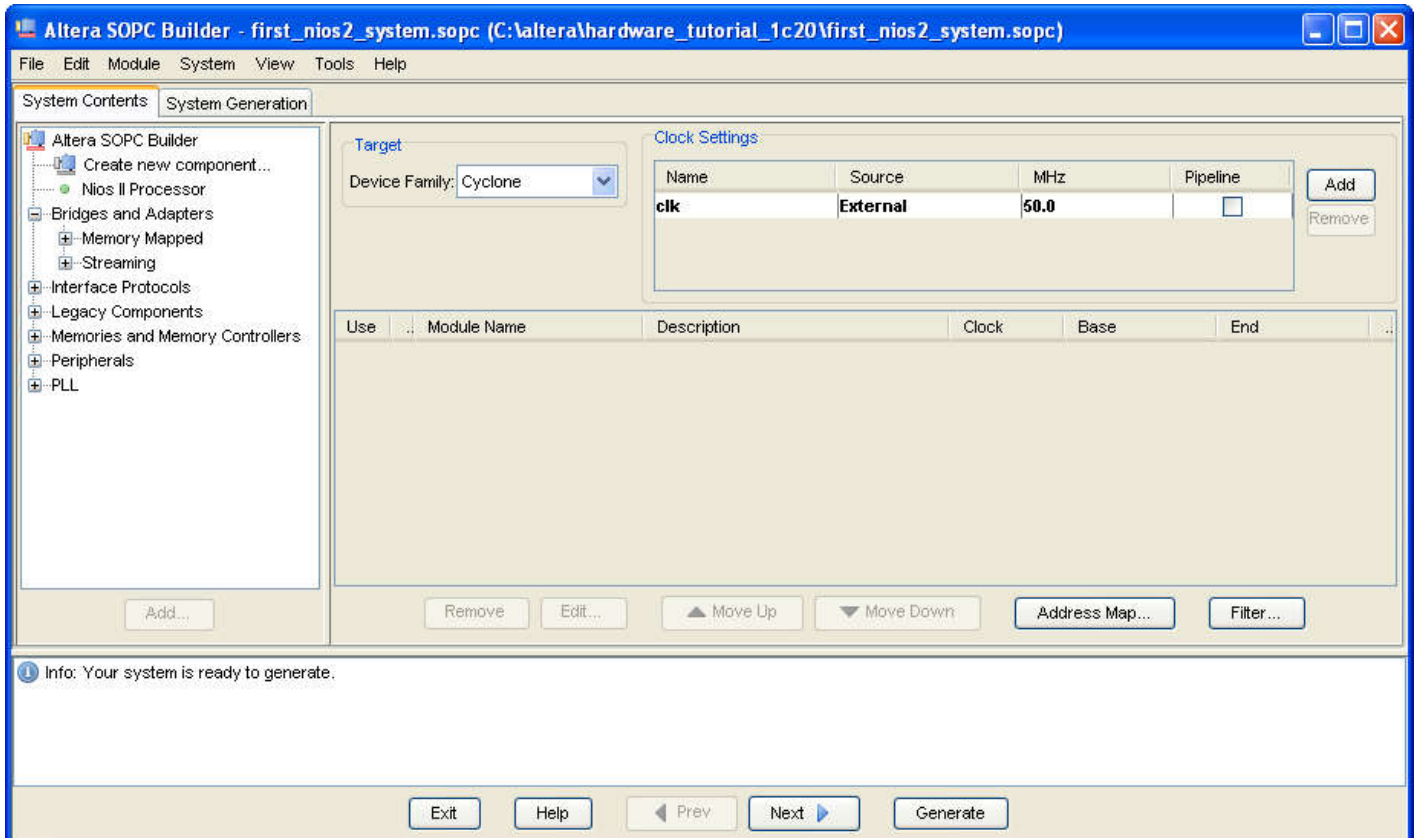
❖ Application requirements:

- What computational performance does the application require?
- How much bandwidth or throughput does the application require?
- What types of interfaces does the application require?
- Does the application require multithreaded software?

❖ System requirements:

- Which Nios II processor core to use: smaller or faster?
- What components does the design require? How many of each kind?
- Which real-time operating system (RTOS) to use, if any?
- Where can hardware acceleration logic dramatically improve system performance? For example:
  - Could adding a DMA component eliminate wasted processor cycles copying data?
  - Could a custom instruction replace the critical loop of a DSP algorithm?
  - Could the Nios II C-to-Hardware (C2H) Acceleration Compiler improve performance?

1. Install the Design Files
2. Analyze System Requirements
3. Start the Quartus II Software and Open the Tutorial Example Design Project
4. Create a New SOPC Builder System
5. Define the System in SOPC Builder
6. Integrate the SOPC Builder System into the Quartus II Project
7. Download Hardware Design to Target FPGA
8. Develop Software Using the Nios IDE
9. Run Program

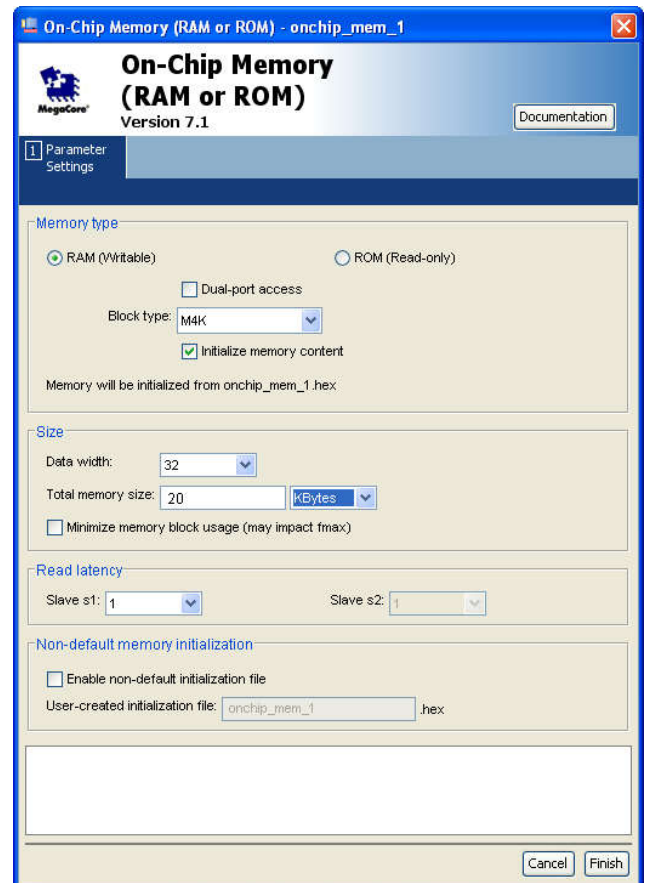


IT-SoC Lab

1. Specify target FPGA and clock settings.
2. Add the Nios II core, on-chip memory, and other components.
3. Specify base addresses and interrupt request (IRQ) priorities.
4. Generate the SOPC Builder system.

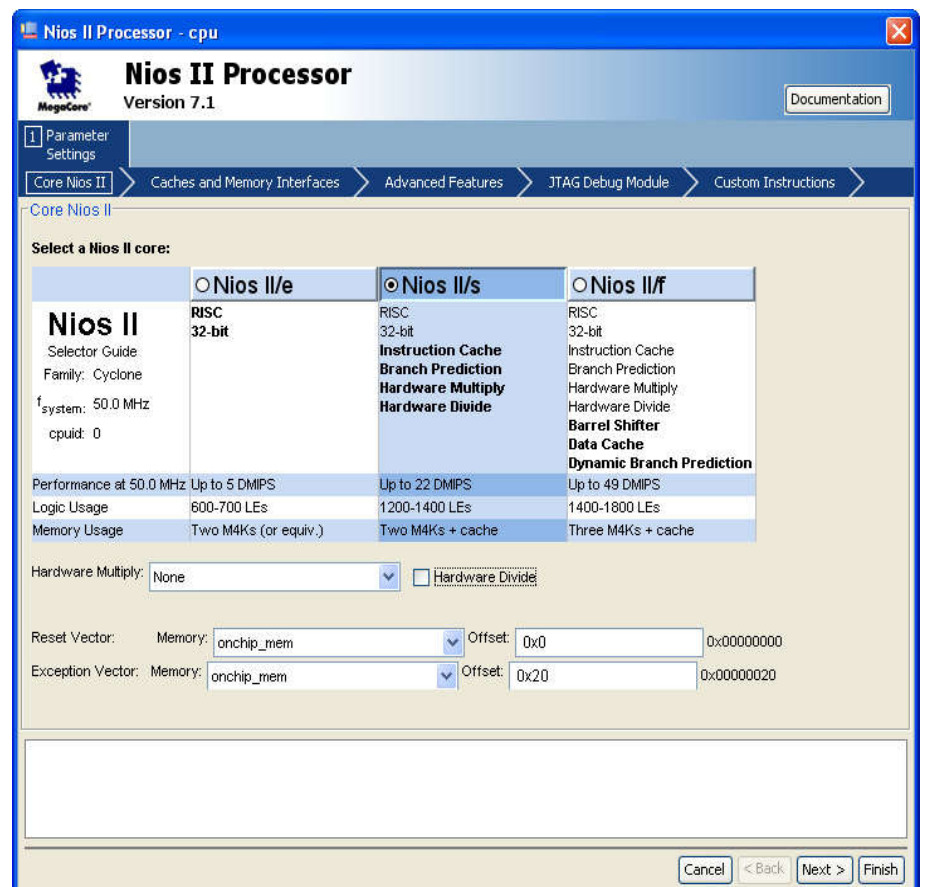
IT-SoC Lab

- ❖ Select **On-chip Memory** from List of available components
- ❖ Click **Add**
- ❖ Select **M4K**
- ❖ Specify **memory size of 20KByte**
- ❖ Click **Finish**



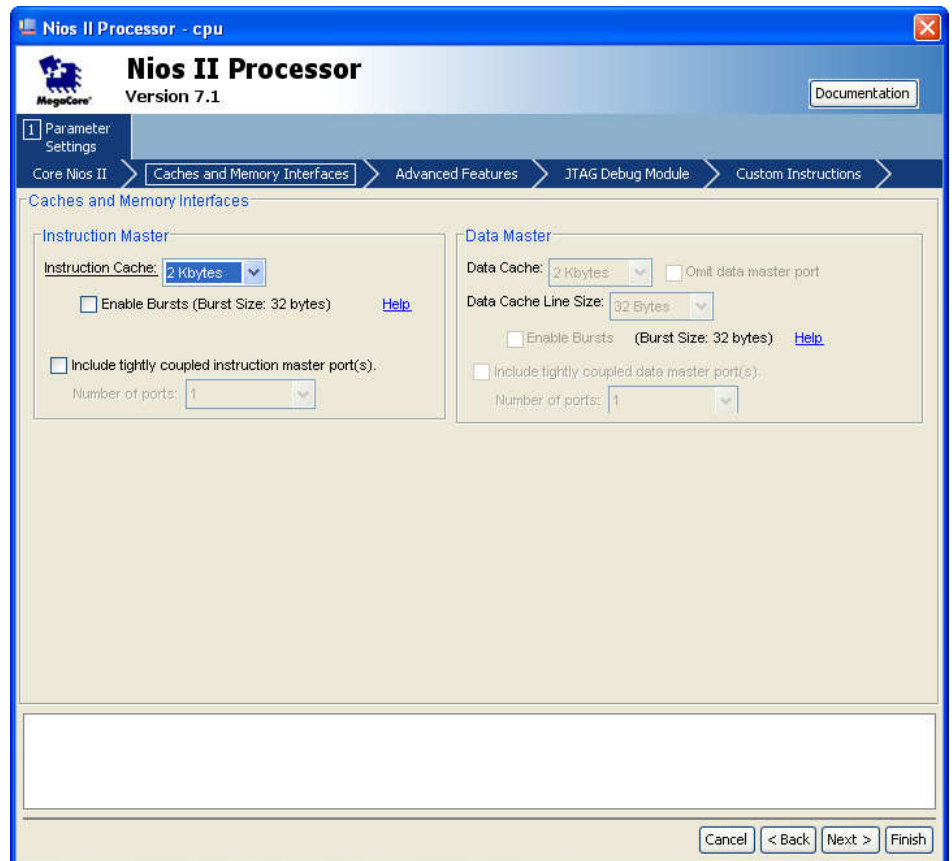
IT-SoC Lab

- ❖ **Nios II Core:** Nios II/s
- ❖ **Hardware Multiply:** None
- ❖ **Hardware Divide:** Off
- ❖ **Reset Vector:** Memory: onchip\_mem Offset: 0x0
- ❖ **Exception Vector:** Memory: onchip\_mem Offset: 0x20



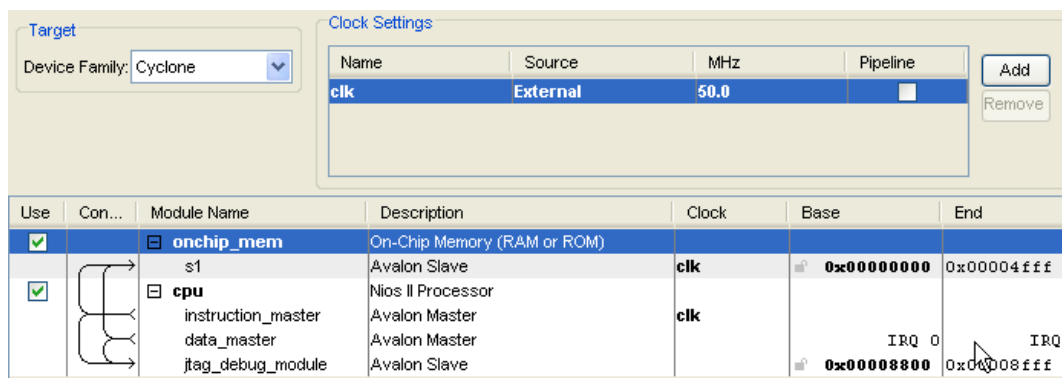
IT-SoC Lab

- ❖ Instruction Cache: 2 Kbytes
- ❖ Enable Bursts: Off
- ❖ Include tightly coupled instruction master port(s): Off



IT-SoC Lab

- ❖ Check the connection



IT-SoC Lab

## ❖ Add the JTAG UART

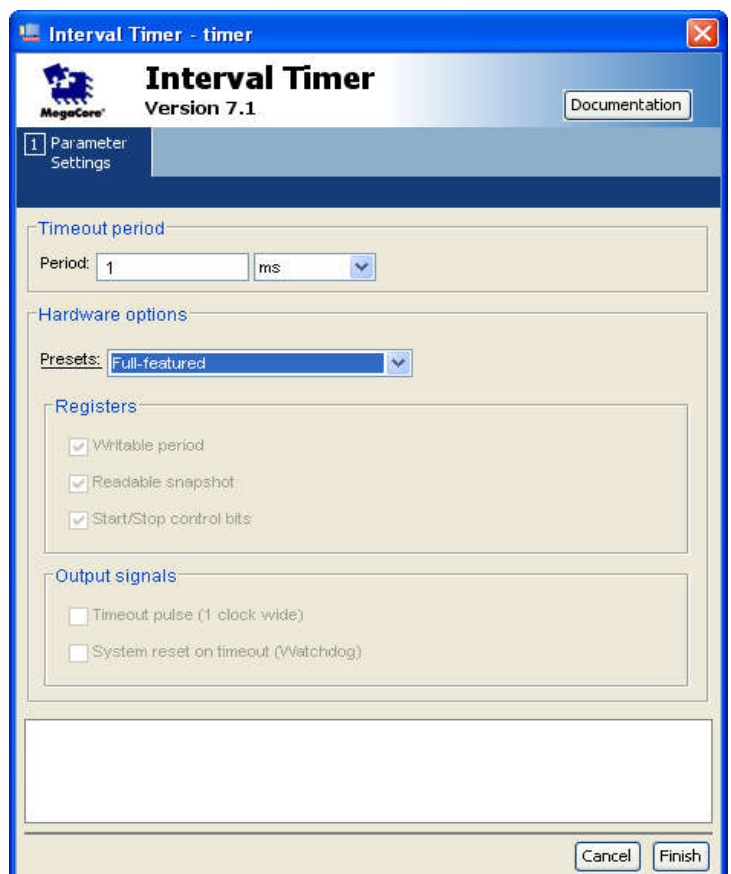
- Select JTAG UART
- Click ADD
- Keep default settings
- Click Finish



IT-SoC Lab

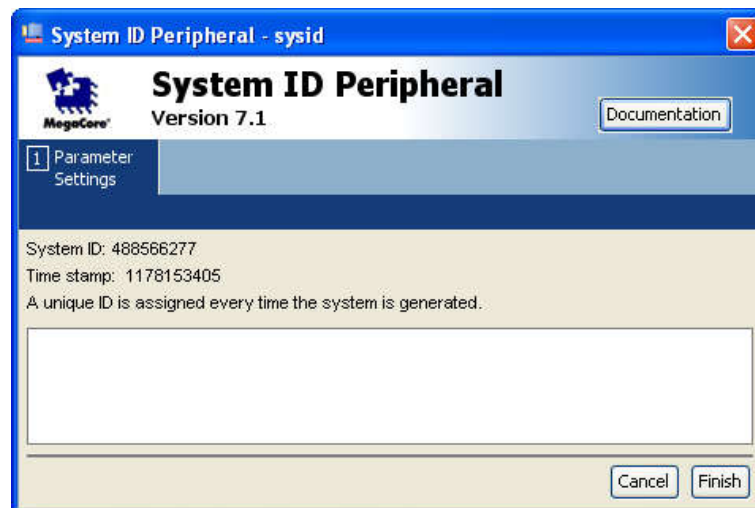
## ❖ Add Interval Timer

- Select Interval Timer from Peripherals list
- Click Add
- Select Full-feature for Presets
- Click Finish



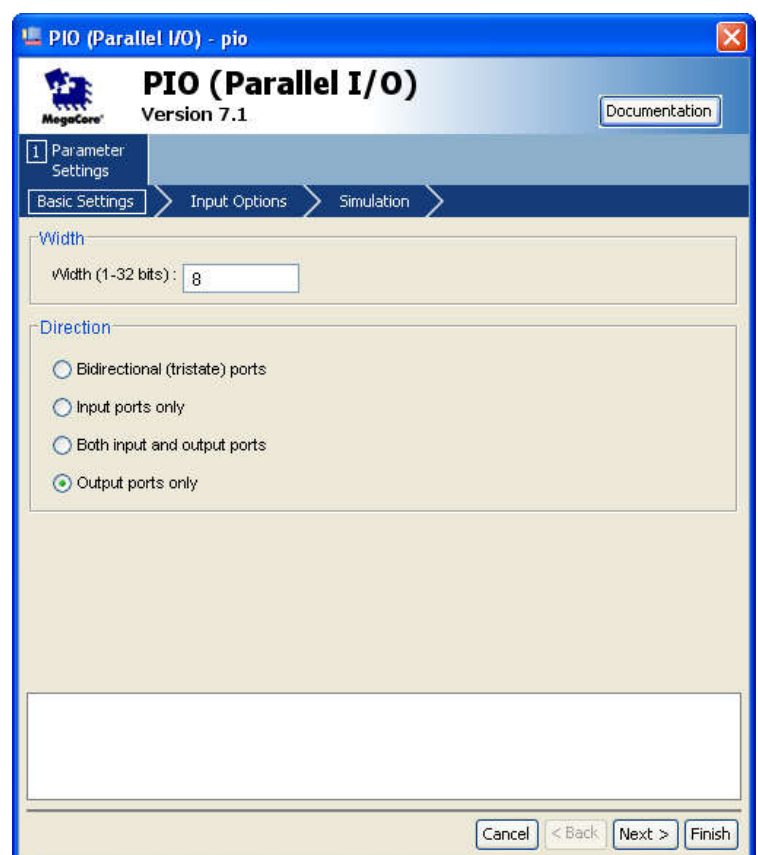
IT-SoC Lab

- ❖ If the system includes the system ID peripheral, the Nios II IDE prevents you from downloading programs compiled for a different system.
- ❖ Perform the following steps to add the system ID peripheral:
  - click System ID Peripheral.
  - Click Add
  - Click Finish



IT-SoC Lab

- ❖ **Select IPO from Peripherals list**
- ❖ **Click Add**
- ❖ **Select Output ports only**
- ❖ **Click Finish**



IT-SoC Lab

- ❖ Click Auto-Assign Base Address
- ❖ Click IRQ
- ❖ Type 16 and press Enter to assign a new IRQ value
- ❖ Turn off Simulation Option

**Target**  
 Device Family: Cyclone

**Clock Settings**

| Name | Source   | MHz  | Pipeline                 |
|------|----------|------|--------------------------|
| clk  | External | 50.0 | <input type="checkbox"/> |

Add  
 Remove

| Use                                 | Con... | Module Name          | Description                 | Clock | Base       | End        | I... |
|-------------------------------------|--------|----------------------|-----------------------------|-------|------------|------------|------|
| <input checked="" type="checkbox"/> |        | <b>onchip_mem</b>    | On-Chip Memory (RAM or ROM) |       |            |            |      |
|                                     |        | s1                   | Avalon Slave                | clk   | 0x00008000 | 0x0000cfff |      |
| <input checked="" type="checkbox"/> |        | <b>cpu</b>           | Nios II Processor           |       |            |            |      |
|                                     |        | instruction_master   | Avalon Master               | clk   |            |            |      |
|                                     |        | data_master          | Avalon Master               |       |            |            |      |
|                                     |        | jtag_debug_module    | Avalon Slave                |       | 0x00010800 | 0x00010fff |      |
| <input checked="" type="checkbox"/> |        | <b>jtag_uart</b>     | JTAG UART                   |       |            |            |      |
|                                     |        | avalon_jtag_slave    | Avalon Slave                | clk   | 0x00011030 | 0x00011037 | 16   |
| <input checked="" type="checkbox"/> |        | <b>sys_clk_timer</b> | Interval Timer              |       |            |            |      |
|                                     |        | s1                   | Avalon Slave                | clk   | 0x00011000 | 0x0001101f | 0    |
| <input checked="" type="checkbox"/> |        | <b>sysid</b>         | System ID Peripheral        |       |            |            |      |
|                                     |        | control_slave        | Avalon Slave                | clk   | 0x00011038 | 0x0001103f |      |
| <input checked="" type="checkbox"/> |        | <b>led_pio</b>       | PIO (Parallel I/O)          |       |            |            |      |
|                                     |        | s1                   | Avalon Slave                | clk   | 0x00011020 | 0x0001102f |      |

IT-SoC Lab

Altera SOPC Builder - first\_nios2\_system.sopc (C:\altera\hardware\_tutorial\_1c20\_design\_files\first\_nios2\_system.sopc)

File Edit Module System View Tools Nios II Help

System Contents System Generation

**Options**  
 System module logic will be created in Verilog.  
☐ Simulation. Create project simulator files. Run Simulator

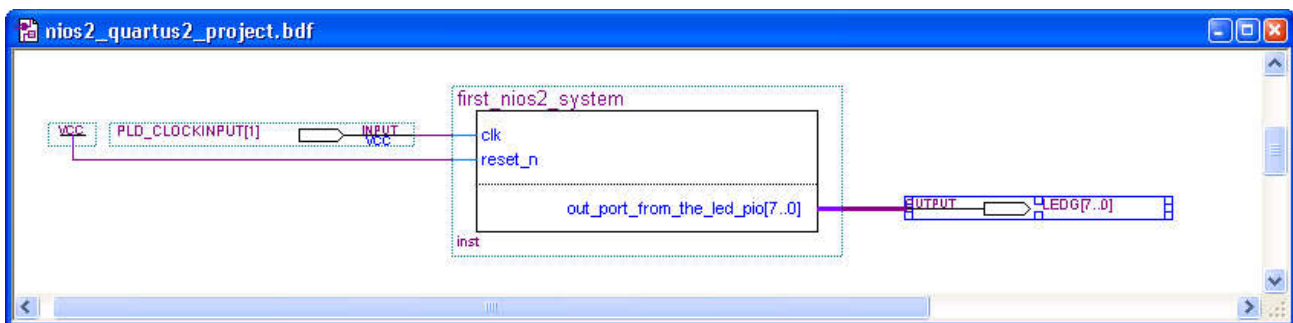
**Nios II Tools**  
Nios II IDE

Info: Altera or its authorized distributors. Please refer to the applicable agreement for further details.  
 Info: Processing started: Wed May 02 17:56:49 2007  
 Info: Command: quartus\_sh -t first\_nios2\_system\_setup\_quartus.tcl  
 Info: Evaluation of Tcl script first\_nios2\_system\_setup\_quartus.tcl was successful  
 Info: Quartus II Shell was successful. 0 errors, 0 warnings  
 Info: Allocated 42 megabytes of memory during processing  
 Info: Processing ended: Wed May 02 17:56:50 2007  
 Info: Elapsed time: 00:00:01  
 # 2007.05.02 17:56:51 (\*) Completed generation for system: first\_nios2\_system.  
 # 2007.05.02 17:56:51 (\*) THE FOLLOWING SYSTEM ITEMS HAVE BEEN GENERATED:  
 SOPC Builder database : C:\altera\hardware\_tutorial\_1c20\_design\_files\first\_nios2\_system.ptf  
 System HDL Model : C:\altera\hardware\_tutorial\_1c20\_design\_files\first\_nios2\_system.v  
 System Generation Script : C:\altera\hardware\_tutorial\_1c20\_design\_files\first\_nios2\_system\_generation\_script  
 # 2007.05.02 17:56:51 (\*) SUCCESS: SYSTEM GENERATION COMPLETED.  
 Info: System generation was successful.  
 Info: Your system is ready to generate.

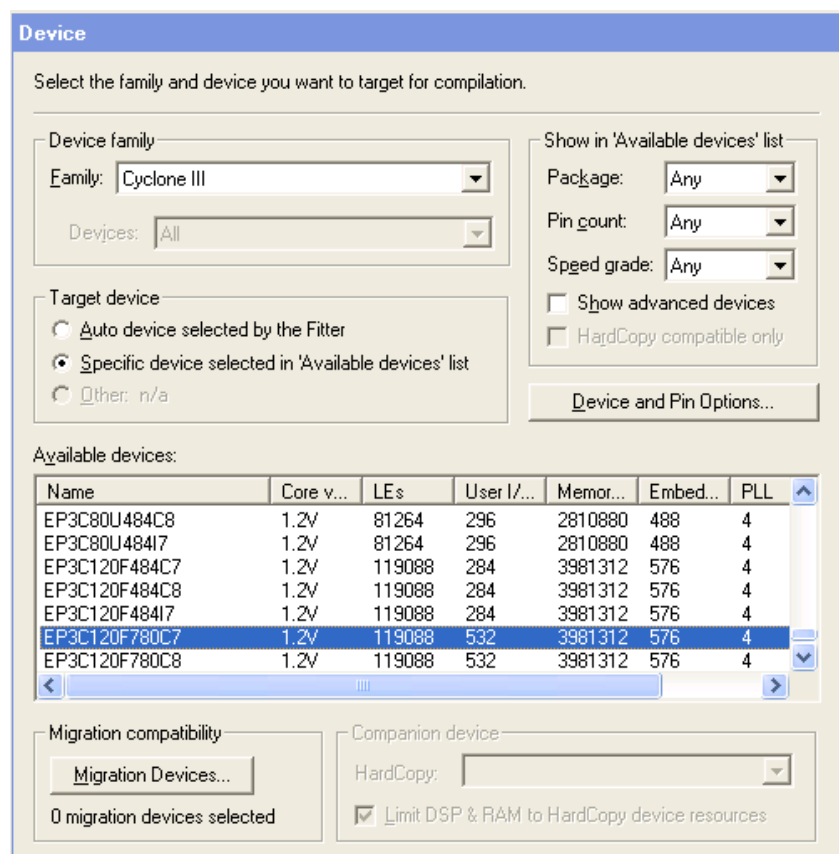
Exit Help Prev Next Generate

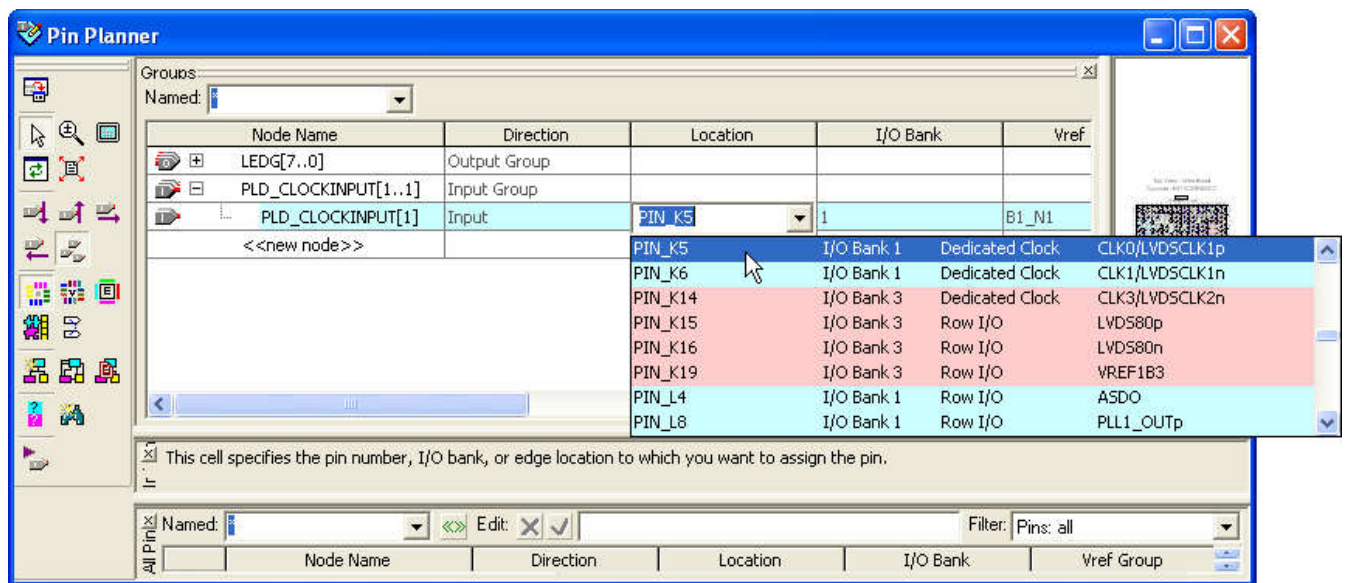
IT-SoC Lab

- ❖ Import the SOPC block into block schematic
- ❖ Add input ports, output ports
- ❖ Make names for input/output ports
- ❖ Provide Vcc to reset\_n signal



- ❖ Select the Cyclone III
  - EP3C120F780C7





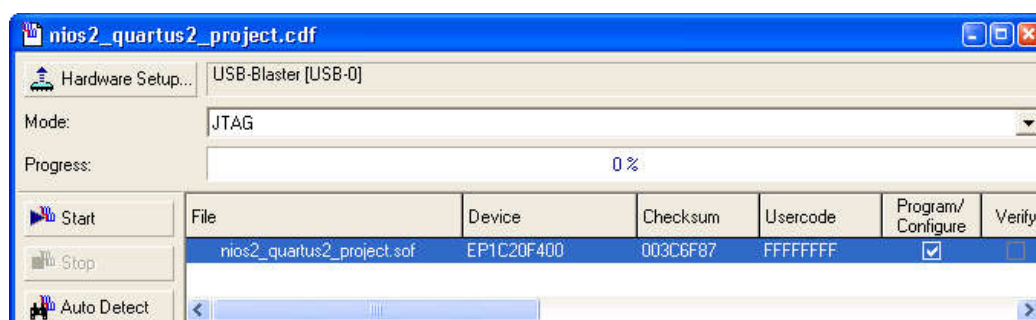
## User LED on Bitec Kit

|     | LED7 | LED6 | LED5 | LED4 | LED3 | LED2 | LED1 | LED0 |
|-----|------|------|------|------|------|------|------|------|
| Pin | AF19 | AG19 | AC17 | AE15 | AD19 | AF18 | AE20 | AD15 |

IT-SoC Lab

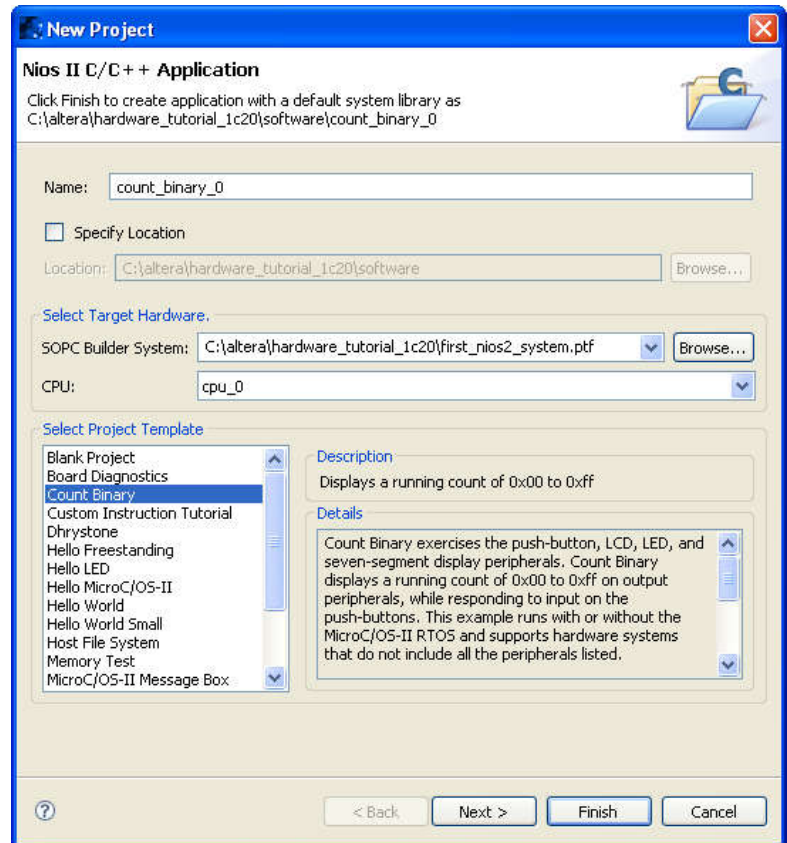
Truong Quang Vinh

- ❖ Connect the board to the host computer with download cable
- ❖ In Quartus software, click Programmer
- ❖ Click Hardware Setup, and select USB-Blaster
- ❖ Select .sof file
- ❖ Click Start



IT-SoC Lab

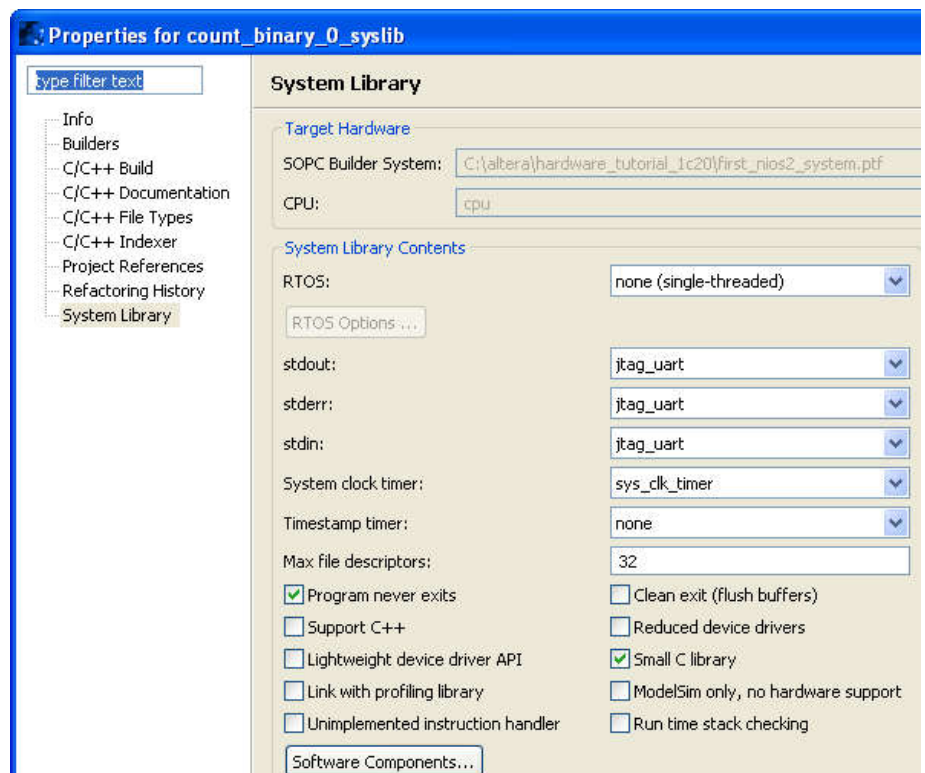
- ❖ Create a new project
- ❖ Select name
- ❖ Select SOPC Builder system file
- ❖ Select Count Binary
- ❖ Click Next



IT-SoC Lab

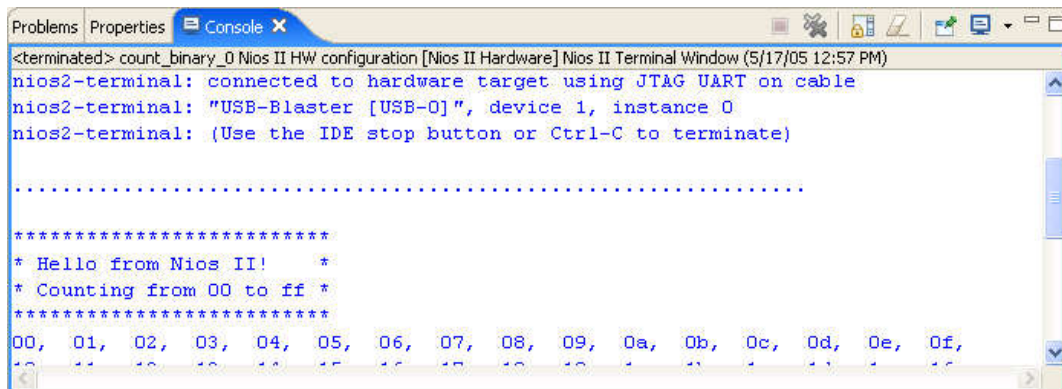
Truong Quang Vinh

- ❖ Turn on Program never exits.
- ❖ Turn off Support C++.
- ❖ Turn off Clean exit (flush buffers).
- ❖ Turn on Small C library.



IT-SoC Lab

- ❖ Right-click the `count_binary_0` project, point to **Run As**, and then click **Nios II Hardware**.
- ❖ Click **Terminate** (the red square) on the toolbar at the upper-right hand corner of the Console view to terminate the run session.



The screenshot shows the Nios II Terminal Window with the following text:

```
<terminated> count_binary_0 Nios II HW configuration [Nios II Hardware] Nios II Terminal Window (5/17/05 12:57 PM)
nios2-terminal: connected to hardware target using JTAG UART on cable
nios2-terminal: "USB-Blaster [USB-0]", device 1, instance 0
nios2-terminal: (Use the IDE stop button or Ctrl-C to terminate)

.....

*****
* Hello from Nios II!      *
* Counting from 00 to ff *
*****
00, 01, 02, 03, 04, 05, 06, 07, 08, 09, 0a, 0b, 0c, 0d, 0e, 0f,
```